No natural number is Dedekind infinite

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Editorial note. The proof's only citation is the Dedekind induction principle, whose defining unit is not among the imported sources; the card injects it as a navigational label so the citation resolves, and the induction itself is carried out in full inside the counted proof.

Editorial note. The Case 2 function h is defined by the author with a displayed two-clause case definition, retained verbatim in the counted span.

Editorial scope. One original proposition of the unit's section on infinity, counted once: the non-Dedekind-infinitude of each natural number. The counted proposition and its author proof are retained in full; the surrounding discussion of the Axiom of Infinity before and after is not counted.

Allowed standard prerequisites (editorial).

- A Dedekind infinite set: a set admitting an injection whose range is a proper subset of its domain; a set is Dedekind infinite when such an injection exists.
- The natural numbers of the iterative construction: each number n is the set of its predecessors, and s(n) = n together with {n} is its successor.
- The Dedekind induction principle: a statement holding of the empty set and passing from n to s(n) holds of every natural number; its defining unit is not among the imported sources, so the card injects it as a navigational label only.
- Restriction of a function to a subset of its domain, and the image (range) and domain notations.

Source reference. Original theorem Dedekind-induction: sources/olp-set-theory-z/pinned-original-source.tex, source lines 540–540. The proof opens by invoking the Dedekind induction principle and then executes the induction in full. It is a navigation pointer to material the counted proof cites; the counted proof retains its own argument in full. The principle's defining unit is not among the imported sources, so the source pointer records the citation site in this unit's own source.

Proposition 0.1. *No natural number is Dedekind infinite.*

Proof. The proof is by induction, i.e., Dedekind-induction . Clearly $0 = \emptyset$ is not Dedekind infinite. For the induction step, we will establish the contrapositive: if (absurdly) $s(n)$ is Dedekind infinite, then n is Dedekind infinite.

So suppose that $s(n)$ is Dedekind infinite, i.e., there is some injection f with $\operatorname{ran} f \subsetneq \operatorname{dom} f = s(n) = n \cup \{n\}$. There are two cases to consider.

Case 1: $n \notin \operatorname{ran} f$. So $\operatorname{ran} f \subseteq n$, and $f(n) \in n$. Let $g = f \restriction n$; now $\operatorname{ran} g = \operatorname{ran} f \setminus \{f(n)\} \subsetneq n = \operatorname{dom} g$. Hence n is Dedekind infinite.

Case 2: $n \in \operatorname{ran} f$. Fix $m \in \operatorname{dom} f \setminus \operatorname{ran} f$, and define a function h with domain $s(n) = n \cup \{n\}$:

$$h(x) = \begin{cases} f(x) & \text{if } f(x) \neq n \\ m & \text{if } f(x) = n \end{cases}$$

So h and f agree everywhere, except that $h(f^{-1}(n)) = m \neq n = f(f^{-1}(n))$. Since f is injection, $n \notin \operatorname{ran} h$; and $\operatorname{ran} h \subsetneq \operatorname{dom} h = s(n)$. Now n is Dedekind infinite, using the argument of Case 1. □